

Template-Sacrificing Synthesis of Well-Defined Asymmetrically-Coordinated Single-Atom Catalysts for Highly Efficient CO₂ Electrocatalytic Reduction

Ming Huang^{1,†}, Bangwei Deng^{2,†}, Xiaoli Zhao², Zheyue Zhang¹, Fei Li³, Kanglu Li⁴, Zhihao Cui⁵, Lingxuan Kong¹, Jianmei Lu⁶, Fan Dong^{2,}, Lili Zhang^{7,*}, and Peng Chen^{1,*}*

¹School of Chemical and Biomedical Engineering, Nanyang Technological University, 70 Nanyang Drive, 637457, Singapore

²Research Center for Environmental Science & Technology, Institute of Fundamental and Frontier Sciences, University of Electronic Science and Technology of China, Chengdu 611731, China

³Material Systems for Nanoelectronics, Chemnitz University of Technology, 09107, Chemnitz, Germany

⁴College of Architecture and Environment, Sichuan University, Chengdu, Sichuan 610065, China

⁵Department of Chemistry and Biochemistry, The Ohio State University, Columbus, Ohio 43210, United States

⁶College of Chemistry, Chemical Engineering and Materials Science, Collaborative Innovation Center of Suzhou Nano Science and Technology, Soochow University, Suzhou 215123, China

⁷Institute of Chemical and Engineering Sciences, A*STAR, 1 Pesek Road, Jurong Island, 627833, Singapore

Corresponding Author

*Peng Chen (ChenPeng@ntu.edu.sg);

*Fan Dong (dongfan@uestc.edu.cn);

*Lili Zhang (zhang_lili@ices.a-star.edu.sg).

ABSTRACT

Although various single-atom catalysts have been designed, atomically engineering their coordination environment remains as a great challenge. Herein, a one-pot template-sacrificing pyrolysis approach is developed to synthesize well-defined Ni-N₄-O catalytic sites on highly porous graphitic carbon for electrocatalytic CO₂ reduction to CO with high Faradaic efficiency (maximum of 97.2%) in a wide potential window (-0.56 to -1.06 V vs. RHE) and high stability. The in-depth experimental and theoretical studies reveal that the axial Ni-O coordination introduces asymmetry to the catalytic center, leading to lower Gibbs free energy for the rate-limiting step, strengthened binding with *COOH, and weaker association with *CO. The present results demonstrate the successful atomic-level coordination environment engineering of high-surface-area porous graphitic carbon-supported Ni single-atom catalysts and the demonstrated method can be applied to synthesize an array of SACs (metal-N₄-O) for various catalysis applications.

KEYWORDS: Ni single-atom catalysts, CO₂ reduction reaction, coordination environment, asymmetry, electronic structure, porous graphitic carbon

Electrocatalytic reduction of CO₂ to energy-rich fuels and value-added industrial chemicals has been recognized as an efficient approach to realize carbon neutrality, and at the same time, store sustainable and renewable electrical energy.¹⁻⁴ However, practical realization of electrocatalytic CO₂ reduction reaction (CO₂RR) is still challenging due to the sluggish reaction kinetics for activation and protonation of chemically inert CO₂ molecules and the competing hydrogen evolution reaction (HER) in aqueous electrolyte.⁵⁻¹⁰ Hence, efficient electrocatalysts for CO₂RR with superior catalytic activity and selectivity are highly demanded.

Single-atom catalysts (SACs) with atomically isolated metal active sites embedded in nitrogen-doped carbon matrix (M-N-C) have attracted tremendous interest owing to ~100% atom utilization efficiency and exceptive electronic structure highly tunable by atomically engineering the coordination environment.¹¹⁻¹⁹ But most M-N-C SACs have a planar porphyrin-like structure (M-N₄) whose symmetric charge distribution is not conducive for electron transfer to the catalytic reactions.^{14,20} Therefore, breaking the symmetry of electronic density by axially introducing an additional coordinating atom shall improve the catalytic performance. A few recent works have demonstrated incorporation of O atom with higher electronegativity than N onto M-N₄ structure to enhance the oxygen reduction reaction and CO₂RR, by oxidizing the synthesized catalyst in air or using oxygen-rich hydrocarbon precursors as both carbon and oxygen source during synthesis.²¹⁻²⁵ But in these syntheses, O atoms can be incorporated into the carbon matrix or consumed by formation of CO₂ or CO at high pyrolysis temperature, making the structure of catalytic sites not well defined and precise formation of abundant M-N₄-O structures with an axial M-O coordination highly challenging.^{12,23-26}

Herein, a template-sacrificing strategy was developed to controllably synthesize Ni-N₄-O SACs anchored on nitrogen-doped porous graphitic carbon (NiSA-N-PGC). During the high-

temperature pyrolysis process, $\text{Mg}(\text{OH})_2$ template decomposes into MgO and water vapor, with the former acting as support for deposition of graphitic carbon and the latter to oxidize Ni atom in form of axial Ni-O coordination. In the presence of highly reactive ammonia gas, N atoms, but not O atoms from water vapor, are doped into the carbon lattice to form planar Ni-N₄ structure. NiSA-N-PGC with abundant well-defined Ni-N₄-O catalytic sites exhibits outstanding performance towards CO₂RR, as evidenced by a Faradic efficiency of CO (FE_{CO}) about 97.2%, a wide operation potential window with $\text{FE}_{\text{CO}} > 90\%$, a turnover frequency (TOF) of 9097 h⁻¹ at -0.76 V, and high stability. Furthermore, comprehensive experimental and theoretical investigations mechanistically reveal the critical roles of axial Ni-O coordination to enhance catalytic activity and selectivity.

RESULTS AND DISCUSSIONS

A simple one-pot template-sacrificing synthesis was developed where $\text{Mg}(\text{OH})_2$ served as the support for carbon deposition, sacrificing porogen and oxygen source, with citric acid as carbon source, ammonia as nitrogen source, and $\text{Ni}(\text{NO}_3)_2$ as Ni precursor (**Figure 1a**). Firstly, the three reactants were mixed and ground into a homogeneous mixture, and subsequently pyrolyzed at 900 °C under NH_3 atmosphere. During the process, $\text{Mg}(\text{OH})_2$ decomposes into MgO and water vapor. The former then serves as the porous support for carbon deposition while the latter oxidizes Ni atoms to form axial Ni-O coordination. After acid leaching, nitrogen-doped porous graphitic carbon anchored with atomically dispersed Ni-N₄-O site (NiSA-N-PGC) was obtained. For comparison, catalysts with various amount of Ni atoms were synthesized denoted as NiSA(*x*)-N-PGC where *x* refers to the mole numbers of added Ni precursor. As shown in scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images, NiSA(0.3)-N-PGC is a highly porous structure formed by interconnected thin-sheets (**Figure 1b, c**). The lattice fringes with characteristic graphitic spacing of 0.34 nm can be clearly resolved under high-resolution TEM

(**Figure 1d**). The aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) shows numerous well-dispersed bright dots, corresponding to heavier Ni atoms on the graphitic carbon matrix (**Figure 1e**). No nanoparticles or clusters were found from SEM, TEM and HAADF-STEM images, suggesting that Ni species are atomically dispersed. Further, energy-dispersive X-ray spectroscopy (EDS) element mapping indicates uniform distribution of C, N, O and Ni elements (**Figure 1f, g**).

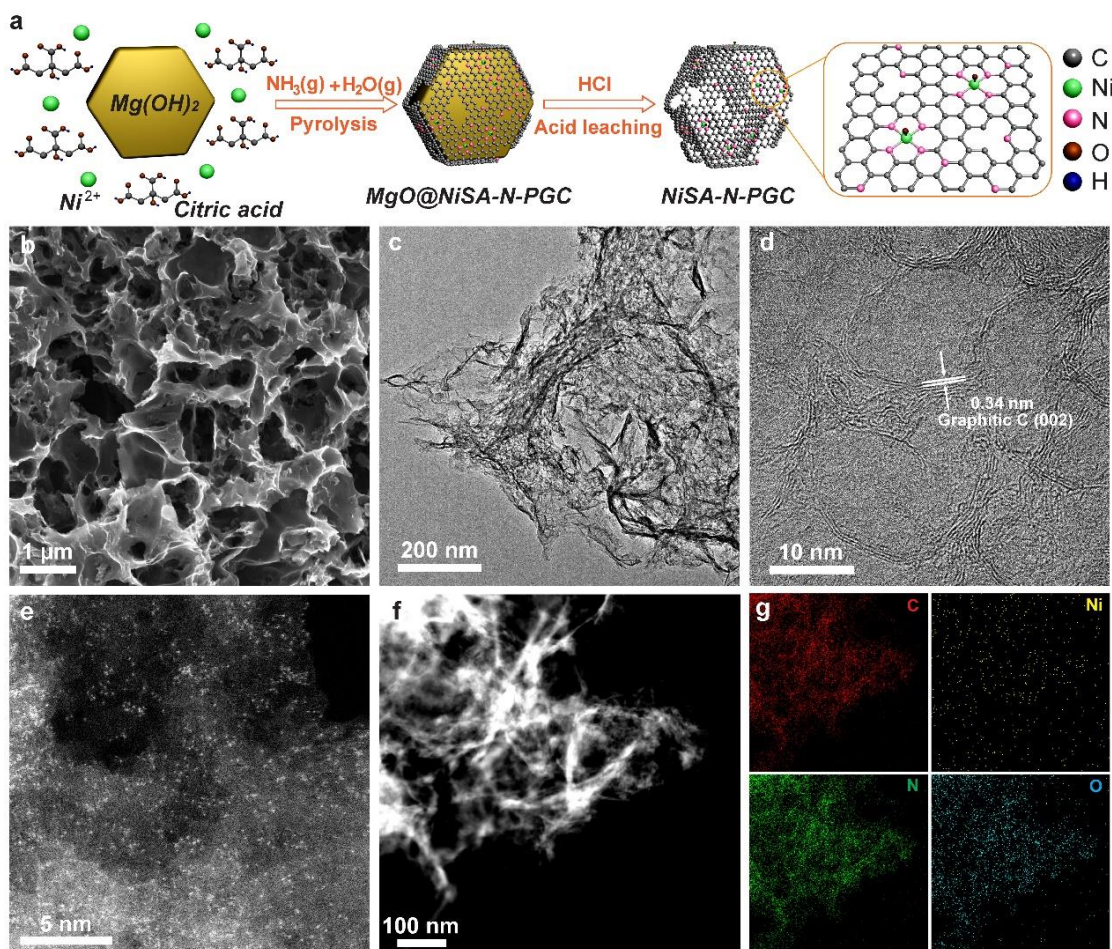


Figure 1. (a) Illustration of the synthesis of nickel single nickel atoms dispersed on nitrogen-doped porous graphitic carbon (NiSA-N-PGC). (b-d) SEM and TEM images of NiSA(0.3)-N-PGC. (e) The aberration-corrected HAADF-STEM image of NiSA(0.3)-N-PGC. (f, g) HAADF-STEM image and corresponding EDS elemental mapping of NiSA(0.3)-N-PGC.

All NiSA-N-PGC samples with various Ni amounts exhibit similar morphology (**Figure S1**), and their Raman spectra show a relatively large intensity ratio between D band and G band ($I_D/I_G \sim 1.1$), indicating a high density of defect sites attributed to N-dopants and embedded Ni single atoms (**Figure S2**). The X-ray diffraction (XRD) patterns of NiSA-N-PGC samples show two broad peaks located at $\sim 26^\circ$ and 44° , attributing to the (002) and (101) diffraction peaks of graphitic carbon, respectively (**Figure 2a, Figure S3**).^{27,28} Nickel or nickel oxide peaks are absent, further indicating the atomic form of Ni species. Note that these two peaks increase with increasing amount of Ni presumably because Ni assists pyrolysis of the carbon precursor. The chemical composition and elemental states of NiSA-N-PGC were investigated by X-ray photoelectron spectroscopy (XPS). The survey XPS spectra show the co-existence of C, N, O, and Ni elements, agreeing with the EDS mapping (**Figure S4**). The high-resolution Cl 2p spectrum of NiSA(0.3)-N-PGC sample shows no Cl residue from the synthesis (**Figure S5**). The intensity of Ni peak is weak due to its low content (**Figure S4a**). The Ni contents in NiSA-N-PGC samples (e.g., 0.51 wt.% for NiSA(0.3)-N-PGC) were measured by inductively coupled plasma optical emission spectroscopy (**Table S1**). The Ni 2p_{3/2} peaks (~ 854.7 eV) in Ni 2p XPS spectra of NiSA-N-PGC samples are located between metallic Ni⁰ peak (852.5-853.0 eV) and Ni²⁺ peak (855.7-856.0 eV), suggesting that the valence of Ni single atom is between 0 and +2 (**Figure S4b**). The high-resolution N 1s XPS spectra are deconvoluted into pyridinic (~ 398.4 eV), Ni-N (~ 399.6 eV), pyrrolic (~ 400.4 eV), graphitic (~ 401.3 eV) and oxidized (~ 402.7 eV) N species (**Figure 2b** and **Figure S4c**).^{14,29} Notably, the Ni-N peak increases with Ni amount (NiSA(0.1)-N-PGC, NiSA(0.2)-N-PGC, and NiSA(0.3)-N-PGC), but saturates at higher amounts (NiSA(0.4)-N-PGC, NiSA(1.0)-N-PGC) presumably because of formation of tiny clusters or nanoparticles (**Figure S6**). In addition, the peak located at 530.5 eV in the O 1s XPS spectra (**Figure 2c**) is assigned to Ni-O

coordination.²⁴ Based on the above analyses, Ni single atoms are individually distributed in N-doped graphitic carbon matrix in the forms of Ni-N and Ni-O coordination. It is noteworthy that Ni-OH shall not exist because OH would decompose into atomic H and O at the high pyrolysis temperature and atomic O then serves as the oxidizing agent to form Ni-O bond.³⁰⁻³²

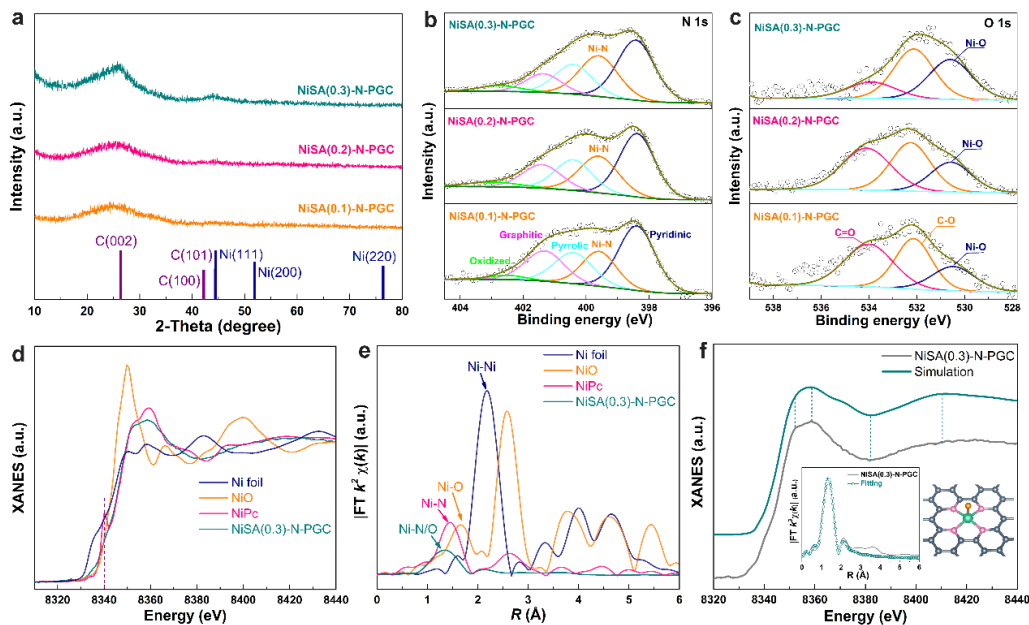


Figure 2. (a) X-ray diffraction patterns of NiSA(0.1)-N-PGC, NiSA(0.2)-N-PGC, and NiSA(0.3)-N-PGC. (b, c) High-resolution N 1s and O 1s spectra of NiSA(0.1)-N-PGC, NiSA(0.2)-N-PGC, and NiSA(0.3)-N-PGC, respectively. (d, e) Ni K-edge XANES spectra (d) and Fourier transformation of the EXAFS spectra (e) of NiSA(0.3)-N-PGC and reference samples. (f) Simulated XANES spectrum of NiSA(0.3)-N-PGC at Ni K-edge. Insets show the corresponding EXAFS fitting curve at R -space and the atomic model of Ni-N₄-O structure in the catalyst (the balls in dark grey, pink, green and brown represent C, N, Ni and O atoms, respectively).

X-ray absorption spectroscopy (XAS) was carried out for the further investigation of the coordination environment of Ni in the catalyst. The Ni K-edge X-ray absorption near-edge structure (XANES) spectrum of NiSA(0.3)-N-PGC shows a near-edge absorption energy between those of Ni foil and NiO, consistent with the XPS observation that the valence state of Ni is between 0 and +2. This is because the coordinating N and O atoms withdraw electrons from Ni. The existence of unpaired electrons of Ni atom was corroborated by electron paramagnetic

resonance (EPR) spectroscopy (**Figure S7**). The g values (e.g., ~ 2.18 for NiSA(0.3)-N-PGC) in the EPR spectra of NiSA-N-PGC samples suggest that the unpaired electrons exist in $3d_{x^2-y^2}$ orbital of Ni.³³ The XAS spectrum of NiSA(0.3)-N-PGC exhibits similar features to those of nickel phthalocyanine (NiPc), implying the existence of Ni-N₄ moieties in NiSA-N-PGC (**Figure 2d**). It's worth noting that the peak of NiPc at ~ 8340 eV is the fingerprint of square-planar Ni-N₄ moiety (D_{4h} centro-symmetry), representing the electron transition from Ni $1s$ to the $4p_z$ orbital.⁸ This peak from NiSA-N-PGC is much weaker due to distortion of D_{4h} symmetry by the additional coordination from O atom.

The Fourier transform (FT) k^3 -weighted $\chi(k)$ function of the extended X-ray absorption fine structure (EXAFS) spectra of NiSA(0.3)-N-PGC and the reference samples are presented in **Figure 2e**. For NiSA(0.3)-N-PGC, the main peak at ~ 1.4 Å could be assigned to the Ni-N/Ni-O scattering path,²⁴ and the peak corresponding to the Ni-Ni scattering path (~ 2.2 Å) is absent. This further confirms that the Ni species are atomically dispersed with coordination with N and O atoms. EXANES wavelet transform (WT) analysis was performed to provide more information in both R and k spaces (**Figure S8**). The WT contour plot of NiSA(0.3)-N-PGC peaks at ~ 4.0 Å⁻¹ is close to that of NiPc and NiO, suggesting the presence of Ni-N/Ni-O coordination. Based on the comprehensive experimental evidences, the atomic model is proposed where a Ni atom is coordinated with four N atoms to form a distorted square-planar Ni-N₄ structure and an additional O atom perpendicularly coordinates with Ni atom *via* an axial Ni-O bond (**Figure S9**). The simulated XANES spectrum of such Ni-N₄-O model based on density functional theory (DFT) calculations and the corresponding Ni K-edge EXAFS fitting curves at R -space perfectly match with the experimental results (**Figure 2f, Figure S10, Table S2**). Our experimental and theoretical investigations ambiguously corroborate the unique Ni-N₄-O coordination configuration, which is

different from the previously reported Ni-N_x catalytic sites³⁴⁻³⁶ and promises an enhanced performance due to symmetry breaking. The Brunauer–Emmett–Teller (BET) surface area and pore volume of NiSA(0.3)-N-PGC are as high as 1133.8 m² g⁻¹ and 1.63 cm³ g⁻¹ (**Figure S11** and **Table S3**), ensuring abundant accessible active sites and unhindered mass transport.³⁷ The vital merit of the presented one-pot synthesis is the use of Mg(OH)₂ template, whose decomposition leads to creation of highly porous structures and *in-situ* generation of water vapor to promote formation of Ni-N₄-O rather than Ni-N₄ sites.

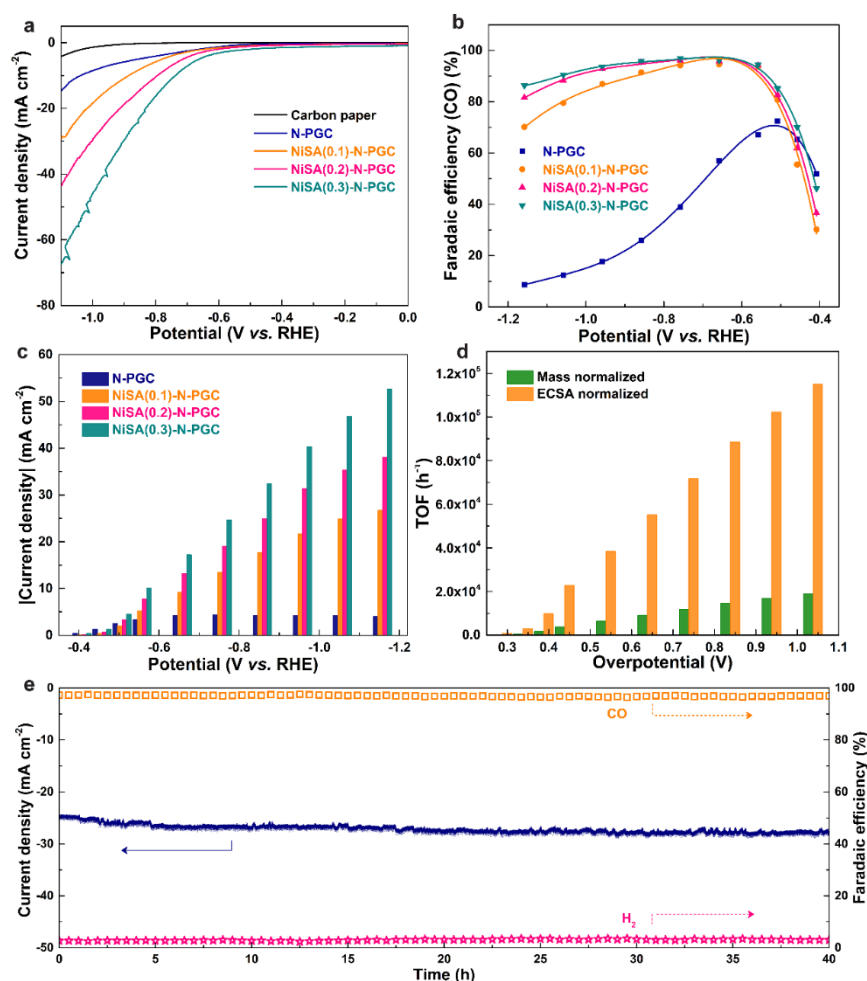


Figure 3. (a) Linear sweep voltammetric (LSV) curves for various catalysts acquired in CO₂-saturated 0.1 M KHCO₃ solution at a scan rate of 5 mV s⁻¹. (b, c) CO Faradaic efficiency and specific current densities of CO for N-PGC, NiSA(0.1)-N-PGC, NiSA(0.2)-N-PGC and NiSA(0.3)-N-PGC at various applied potentials. (d) TOF of NiSA(0.3)-N-PGC at different overpotentials. (e) Long-term stability test of NiSA(0.3)-N-PGC for 40 hours at -0.76 V.

We evaluated the electrocatalytic CO₂RR performance of NiSA-N-PGC catalysts in a CO₂-saturated 0.1 M KHCO₃ solution using a H-type three-electrode cell. The gaseous and liquid products were analyzed by on-line gas chromatography (GC, **Figure S12**) and off-line ¹H nuclear magnetic resonance (¹H NMR) measurements (**Figure S13**). The results show that CO and H₂ are the dominant reduction products, and no liquid product is detected. Carbon paper and nitrogen-doped graphitic carbon (N-PGC) show small current responses in linear sweep voltammetry (**Figure 3a**), indicating a poor CO₂ reduction activity. Anchoring Ni single atoms onto N-PGC results in a significant increase of current density. NiSA(0.3)-N-PGC offers a current density of 67 mA cm⁻² at -1.1 V vs. RHE, outperforming NiSA(0.1)-N-PGC and NiSA(0.2)-N-PGC because of more abundant Ni catalytic centers. However, further increase of Ni content (NiSA(0.4)-N-PGC and NiSA(1.0)-N-PGC) gives negligible improvement (**Figure S14**).

Chronoamperometry curve of NiSA(0.3)-N-PGC was recorded at different potentials and accompanied by GC analysis of CO₂RR products, in order to determine Faradaic efficiency (**Figure S15**). **Figure 3b** presents the Faradaic efficiencies for CO production (FE_{CO}) at various potentials for different catalysts. NiSA(0.3)-N-PGC gives the maximum FE_{CO} of 97.2% at -0.76 V vs. RHE (overpotential = 640 mV) which is higher than that of NiSA(0.2)-N-PGC (96.1%) and NiSA(0.1)-N-PGC (94.6%), and far surpassing that of N-PGC (72.4%) (**Figure S16**). While further increase of Ni content (NiSA(0.4)-N-PGC and NiSA(1.0)-N-PGC) leads to decrease of maximum FE_{CO} (95.6% and 94.7%). This is likely due to the formation of Ni clusters or nanoparticles (**Figure S6**), in which Ni-Ni bonds favors the adsorption and activation of H⁺, leading to enhancement of competitive hydrogen evolution reaction (HER) and thus impairment of the overall performance for CO₂RR.^{38,39} The maximum FE of H₂ production from the competing HER for NiSA(0.3)-N-PGC is <3%. This result suggests that a small amount of Ni could greatly

improve the selectivity of CO formation, highlighting the superiority of Ni SACs for CO₂RR. For comparison, the oxygen-free single-atom Ni catalyst with symmetric N₄ coordination (Ni-N₄-PGC) was also synthesized using MgO as the sacrificing template (Supporting Information, **Figure S17** and **S18**). Notably, Ni-N₄-PGC catalyst delivered smaller CO partial current densities and FE_{CO} at all applied potentials as compared to NiSAC(0.3)-N-PGC. These results demonstrate the key role of the axial Ni-O coordination to enhance CO₂RR. In addition, FE_{CO} of NiSA(0.3)-N-PGC can be maintained >90% from -0.56 to -1.06 V, demonstrating a much wider operation potential window as compared to most transition metal SACs (**Table S4**).^{14,28,40-43} At ~1.16 V, NiSA(0.3)-N-PGC achieves the highest CO partial current density (j_{CO}) of 52.7 mA cm⁻², which is about 13 and 2 times of that of N-PGC and NiSA(0.1)-N-PGC, respectively, demonstrating the high catalytic activity towards CO₂RR (**Figure 3c**).

The Tafel slope of NiSA(0.3)-N-PGC catalysis is ~115 mV dec⁻¹ which is close to the theoretical value (118 mV dec⁻¹) of the initial one-electron transfer step (**Figure S19**), implying that generation of *COOH intermediate on the catalyst surface is the rate-determining step.^{6,44-46} This value is smaller than that of the other Ni SACs, suggesting the lowest energy barrier for *COOH generation on NiSA(0.3)-N-PGC and thus explaining its highest FE_{CO} and j_{CO} . The turnover frequency (TOF) for CO₂RR normalized to total Ni sites and electrochemical active surface area (ECSA) were calculated based on the total mass loading of Ni on the electrode and the measured electrochemical double layer capacitance (EDLC), respectively (**Figure 3d** and **Figure S20-21**). At an overpotential of 640 mV, TOFs of NiSA(0.3)-N-PGC normalized by Ni sites and ECSA are 9097 and 55150 h⁻¹, respectively, which are much higher than that of previously reported NiSA catalysts (**Table S4**).^{20,34,47-49} Furthermore, the long-term stability of both catalytic activity and efficiency is excellent (**Figure 3e**). There was no apparent alteration of

the morphology and chemical composition of Ni-N₄-O catalytic center in NiSA(0.3)-N-PGC catalyst after 40 h electrocatalysis, further demonstrating its outstanding stability (**Figure S22**).

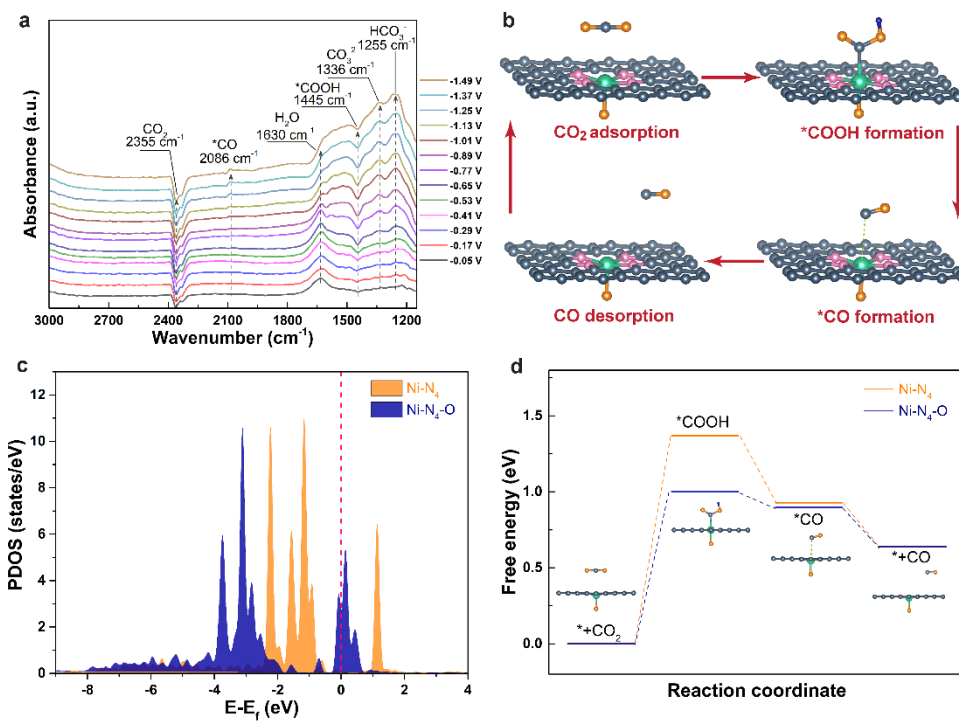


Figure 4. (a) Potential-dependent operando ATR-SEIRAS spectra of NiSA(0.3)-N-PGC in CO₂-saturated 0.1 M KHCO₃ electrolyte. (b) Proposed reaction steps of electrocatalytic reduction of CO₂ to CO over Ni-N₄-O catalytic site. (c) Projected density of states (PDOS) calculated by DFT for Ni 3d of Ni-N₄ and Ni-N₄-O catalytic sites. (d) Reaction paths and free energy diagrams of CO₂ reduction to CO over Ni-N₄ and Ni-N₄-O catalytic sites. Inset shows the side view of the optimized configurations of intermediates over the Ni-N₄-O catalytic site.

To identify the reaction intermediates absorbed on the catalyst surface and understand the catalytic mechanism, operando attenuated total reflection surface enhanced infrared absorption spectroscopy (ATR-SEIRAS) was performed at different potentials with a step interval of 0.12 V in CO₂-saturated 0.1 M KHCO₃ solution. As shown in **Figure 4a**, several vibrational bands are observed while stepping the potential from -0.05 to -1.49 V. Two apparent peaks at 2355 and 1630 cm⁻¹ are attributable to adsorbed CO₂ and H₂O, respectively. With the potential sweeping to more negative levels, the intensity of these two peaks decreases simultaneously due to continuous

consumption of CO₂ and H₂O molecules.⁵⁰ The peaks at 1445 and 2086 cm⁻¹ could be assigned to *COOH and *CO, which are the main reaction intermediates during CO₂RR. Notably, the intensity of *CO peak increases as overpotential shifts to more negative levels, suggesting enhanced CO₂RR activity. This is accompanied by decrease of *COOH peak (hence increase of *COOH consumption), indicating rapid protonation of *COOH to form *CO species.

The electrolytic reduction of CO₂ to CO includes two proton/electron transfer steps: i) activation and protonation of CO₂ molecule to form the first intermediate *COOH (*+ CO₂ + H⁺ + e⁻ → *COOH); and ii) formation of *CO via further hydrogenation of *COOH (*COOH + H⁺ + e⁻ → *CO + H₂O). Finally, *CO desorbs from the catalyst surface to produce CO gas (*CO → * + CO(g)). This process on Ni-N₄-O is illustrated in **Figure 4b**. In order to reveal the key role of axial Ni-O coordination of Ni-N₄-O catalytic site for CO₂RR, density functional theory (DFT) calculations were performed. It's worth noting that the grand canonical DFT method has been used to study substantial effects of surface charges induced by the applied potentials on the electrochemical reactions,⁵¹⁻⁵³ but this method requires an accurate value of the work function of standard hydrogen electrode (SHE) to match corresponding experimental electrode potentials. Thus, we used the most commonly adopted method (computational hydrogen electrode (CHE) approach)⁵⁴ to simulate concerted proton–electron transfers and estimate the adsorption energies of intermediates for elementary steps. Like all the theoretical studies, our approach is based on simplified assumptions but without losing the generality. The structure is optimized *via* relaxing all the atoms in our model until the minimum energy is achieved (**Note S1**). We have also confirmed that each adsorbed intermediate is in the true minimum with real vibrational frequency. As shown in **Table S5**, the Ni center exhibits a higher oxidation state in Ni-N₄-O (+1.12 e) than that in Ni-N₄ catalytic site (+0.91 e). This is due to electron transfer from Ni to the axial O as

clearly evidenced by comparing the charge density difference plots of Ni-N₄ and Ni-N₄-O (**Figure S23**). Further analysis of the projected density of states (PDOS) shows that Ni 3d orbital of Ni-N₄-O exhibits much higher DOS as compared to Ni-N₄ near the Fermi level (**Figure 4c**). Electron delocalization of Ni catalytic center because of larger electronegativity of O atom shall significantly influence the reaction energy barrier and adsorption energy.

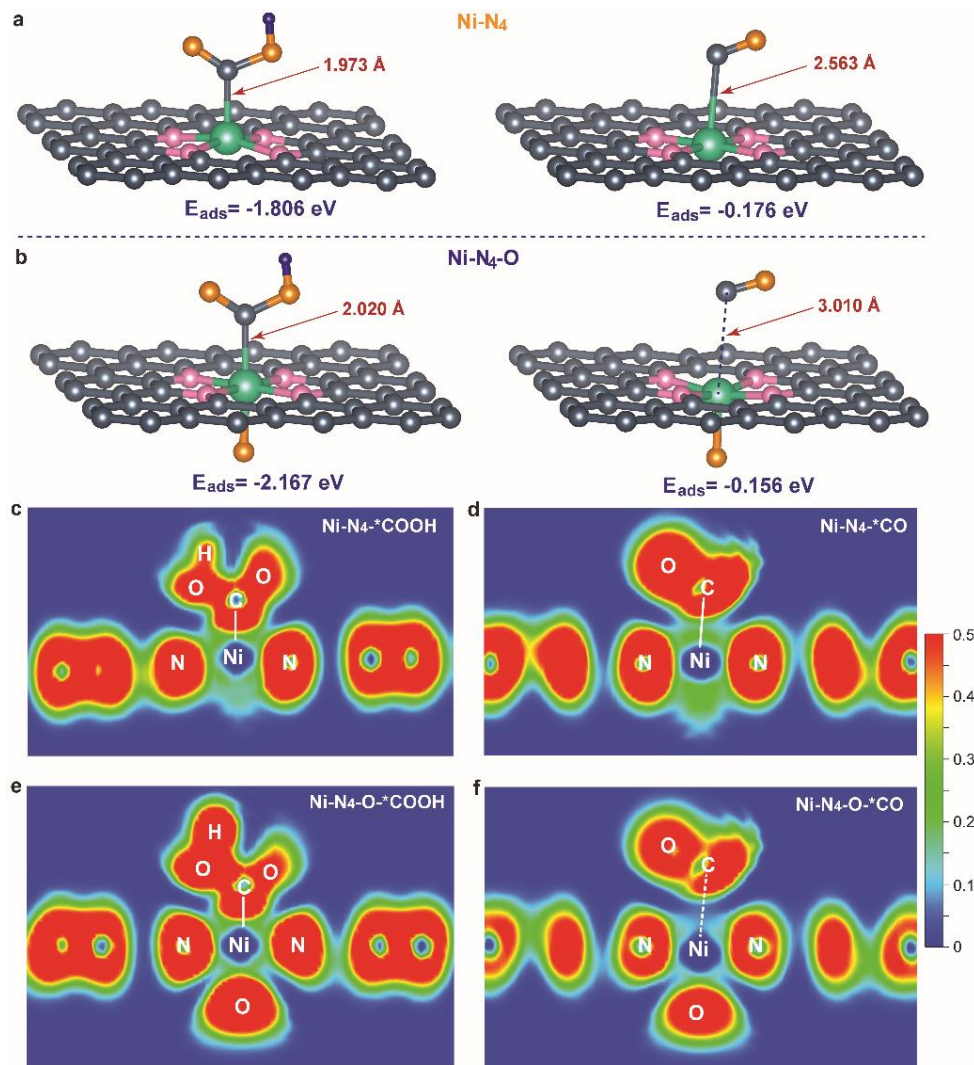


Figure 5. (a) The DFT-optimized configurations of *COOH and *CO adsorbed on Ni-N₄ structure. (b) The DFT-optimized configurations of *COOH and *CO adsorbed on Ni-N₄-O structure. (c, d) Electron localization function (ELF) of *COOH and *CO on Ni-N₄ catalytic site. (e, f) Electron localization function (ELF) of *COOH and *CO on Ni-N₄-O catalytic site.

As shown in the calculated free energy diagram (**Figure 4d, Table S6**), the first protonation of CO₂ to form *COOH is the rate-limiting step for both Ni-N₄ and Ni-N₄-O catalytic sites. The free-energy change (ΔG) for this step on Ni-N₄-O (1.002 eV) is significantly smaller than that on Ni-N₄ (1.37 eV). And the adsorption energy of *COOH on Ni-N₄-O (-2.167 eV) is stronger than that on Ni-N₄ (-1.806 eV), favoring its transformation into *CO due to the axial Ni-O coordination (**Figure 5a and b**). A solvation correction of -0.38 eV to *COOH was applied due to the presence of OH group, based on the explicit solvent model described elsewhere.^{55,56} On the other hand, the relatively weaker binding strength of *CO (E_{ads} = -0.156 vs. -0.176 eV) and larger distance between absorbed *CO and Ni center on Ni-N₄-O (3.010 vs. 2.563 Å) are favorable for desorption of *CO. Electron localization function (ELF) maps confirm that Ni-N₄-O has a strong bonding with *COOH but a weak interaction with *CO (**Figure 5c-f**). Additionally, the interaction between *CO and the catalyst surface can be well interpreted by the difference of PDOS of Ni 3d orbitals of Ni-N₄ and Ni-N₄-O after the absorption of *CO. The d-band center of Ni 3d orbital in Ni-N₄-O (-3.036 eV) is much lower than that in Ni-N₄ (-2.041 eV), explaining the weaker adsorption strength of *CO on Ni-N₄-O (**Figure S24**).^{25,57} Altogether, these theoretical investigations demonstrate that addition of axial Ni-O coordination enhances electronic delocalization of Ni catalytic center whereby lowering the energy barrier of the rate-limiting step, strengthening *COOH adsorption, and promoting *CO desorption.

CONCLUSIONS

In summary, we developed a facile and low-cost strategy to construct Ni single atom catalyst coordinated laterally by four N atoms and axially by one O atom for high-performance CO₂RR. Mg(OH)₂ was used as the sacrificing template to support carbon deposition, enables creation of a highly porous structure after leaching, and serve as oxygen source. This strategy can

be applied to synthesize various asymmetrically-coordinated single-atom-catalysts (M-N₄-O, M = Cu, Co, Fe, Zn, Mn, Pd, Cd, etc.) by choosing different metal hydroxides (such as Ca(OH)₂, Ba(OH)₂, and Sr(OH)₂) as the sacrificing templates to enrich the family of SACs for various catalytic applications. We also demonstrate the superiority of SAC over nanoparticle catalyst and the necessity to optimize SAC loading. Based on comprehensive experimental and theoretical investigations, we show that the axial Ni-O greatly enhances the electronic delocalization of Ni center because of high electronegativity of O and the induced asymmetric distribution of charges promote their participation in the reactions. The high catalytic activity, selectivity and stability of NiSA-N-PGC catalyst are ascribed to low energy barrier for *COOH formation, strong binding strength with *COOH, weak binding with *CO, unhindered mass transport, high specific surface area, and interconnected conductive network. This study provides insights to devise CO₂RR electrocatalysts and the catalytic mechanisms.

METHODS

Chemicals

All chemicals including magnesium hydroxide (95%), citric acid ($\geq 99.5\%$), nickel(II) nitrate hexahydrate (99.999%), KHCO₃ (99.7%) and hydrochloric acid (37%) were purchased from Sigma-Aldrich and directly used without purification.

Synthesis of NiSA-N-PGC catalysts and Ni-N₄-PGC catalyst

5 g citric acid (C₆H₈O₇), various amounts (0.1, 0.2, 0.3, 0.4, and 1.0 mmol) of Ni(NO₃)₂·6H₂O and 5 g magnesium hydroxide (Mg(OH)₂) were ground into homogeneous powder. It was then loaded into a tube furnace and heated at 900 °C for 2 h (ramping rate: 3 °C min⁻¹) under a mixed gas flow of 100 sccm N₂ and 100 sccm NH₃. After heat-treatment, the resulting powder was acid leached

(3M HCl) at 80 °C for 24 h. Afterwards, vacuum filtration and freeze-drying were performed to obtain the final products (NiSA(0.1)-N-PGC, NiSA(0.2)-N-PGC, NiSA(0.3)-N-PGC, NiSA(0.4)-N-PGC, and NiSA(1.0)-N-PGC). The production yield of NiSA-N-PGC is ~15 wt.%. For comparison, nitrogen-doped porous graphitic carbon (N-PGC) was prepared using the same procedure but without adding $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$. To synthesize O-free NiSAC catalyst for comparison (Ni-N₄-PGC), Firstly, $\text{Mg}(\text{OH})_2$ powder was annealed at 800 °C for 5 h (ramping rate: 5 °C min⁻¹) to obtain magnesium oxide (MgO) powder. Using the same synthetic route for NiSA-N-PGC, but replacing $\text{Mg}(\text{OH})_2$ with the same amount of MgO, Ni-N₄-PGC was obtained.

Characterizations

The crystallinity and composition of the synthesized samples were characterized by X-ray diffraction (XRD; Rigaku Co.) with Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$) and X-ray photoelectron spectroscopy (XPS; ESCALAB250Xi, Thermo Fisher Scientific). The morphological features and detailed structural information were conducted by field-emission scanning electron microscopy (FESEM; GeminiSEM, Zeiss), and transmission electron microscopy (TEM; JEOL JEM-2100F). Raman spectroscopy was recorded using a WITEC CRM200 confocal Raman spectrometer (532 nm laser). High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) was carried out by a JEOL JEMARM200F STEM/TEM (resolution of 0.08 nm). Electron paramagnetic resonance (EPR) was investigated by a JES FA200 spectrometer. The X-ray absorption fine structure (XAFS) spectra were recorded at BL14W1 beamline of Shanghai Synchrotron Radiation Facility (SSRF). The Ni concentrations in the NiSA-N-PGC catalysts were measured by the inductively coupled plasma-atomic emission spectroscopy (ICP-AES, SPECTRO-BLUE).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/XXXXX>.

Experimental details of sample preparation and characterizations, SEM, Raman, XRD, XPS, HRTEM, EPR, Nitrogen adsorption-desorption isotherms and EXAFS characterization data of as-prepared catalysts, detailed electrochemical tests and DFT calculations (PDF)

Author contributions

[†]M.H. and B.D. contributed equally to this work. M.H. and P.C. conceived and designed the research. M.H. performed all the experimental and theoretical calculations. B.D and X.Z helped with the electrochemical tests. Z.Z., F.L., K.L., Z.C., L.K., and J.L. contributed to the data analysis and discussions. M.H. wrote and revised the manuscript under the supervision of F.D., L.Z., and P.C. All authors read and approved the final manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENT

This work was supported by an AME-IRG grant (A1983c0025) from Agency for Science, Technology and Research (A*STAR) of Singapore, an AcRF tier 2 grant (MOE2017-T2-2-005) from Ministry of Education (Singapore), the National Natural Science Foundation of China (21822601 and 22176029), the Sichuan Natural Science Foundation for Distinguished Scholars (2021JDJQ0006), and the Basic Research Project of leading Technology in Jiangsu Province (BK20202012).

REFERENCES

- (1) Zhuang, T.-T.; Liang, Z.-Q.; Seifitokaldani, A.; Li, Y.; De Luna, P.; Burdyny, T.; Che, F.; Meng, F.; Min, Y.; Quintero-Bermudez, R.; Dinh, C. T.; Pang, Y.; Zhong, M.; Zhang, B.; Li, J.; Chen, P.-N.; Zheng, X.-L.; Liang, H.; Ge, W.-N.; Ye, B.-J.; Sinton, D.; Yu, S.-H.; Sargent, E. H. Steering Post-C–C Coupling Selectivity Enables High Efficiency Electroreduction of Carbon Dioxide to Multi-Carbon Alcohols. *Nat. Catal.* **2018**, *1*, 421-428.
- (2) Garg, S.; Li, M.; Weber, A. Z.; Ge, L.; Li, L.; Rudolph, V.; Wang, G.; Rufford, T. E. Advances and Challenges in Electrochemical CO₂ Reduction Processes: An Engineering and Design Perspective Looking Beyond New Catalyst Materials. *J. Mater. Chem. A* **2020**, *8*, 1511-1544.
- (3) Qiao, J.; Liu, Y.; Hong, F.; Zhang, J. A Review of Catalysts for the Electroreduction of Carbon Dioxide to Produce Low-Carbon Fuels. *Chem. Soc. Rev.* **2014**, *43*, 631-675.
- (4) Duan, X.; Xu, J.; Wei, Z.; Ma, J.; Guo, S.; Wang, S.; Liu, H.; Dou, S. Metal-Free Carbon Materials for CO₂ Electrochemical Reduction. *Adv. Mater.* **2017**, *29*, 1701784.
- (5) Zhang, Y.-J.; Sethuraman, V.; Michalsky, R.; Peterson, A. A. Competition between CO₂ Reduction and H₂ Evolution on Transition-Metal Electrocatalysts. *ACS Catal.* **2014**, *4*, 3742-3748.
- (6) Zhu, D. D.; Liu, J. L.; Qiao, S. Z. Recent Advances in Inorganic Heterogeneous Electrocatalysts for Reduction of Carbon Dioxide. *Adv. Mater.* **2016**, *28*, 3423-3452.
- (7) Zhang, L.; Zhao, Z.-J.; Gong, J. Nanostructured Materials for Heterogeneous Electrocatalytic CO₂ Reduction and their Related Reaction Mechanisms. *Angew. Chem., Int. Ed.* **2017**, *56*, 11326-11353.
- (8) Colpas, G. J.; Maroney, M. J.; Bagyinka, C.; Kumar, M.; Willis, W. S.; Suib, S. L.; Mascharak, P. K.; Baidya, N. X-Ray Spectroscopic Studies of Nickel Complexes, with Application to the Structure of Nickel Sites in Hydrogenases. *Inorg. Chem.* **1991**, *30*, 920-928.
- (9) Wang, S.; Zhou, P.; Zhou, L.; Lv, F.; Sun, Y.; Zhang, Q.; Gu, L.; Yang, H.; Guo, S. A Unique Gas-Migration, Trapping, and Emitting Strategy for High-Loading Single Atomic Cd Sites for Carbon Dioxide Electroreduction. *Nano Lett.* **2021**, *21*, 4262-4269.
- (10) Ni, W.; Liu, Z.; Zhang, Y.; Ma, C.; Deng, H.; Zhang, S.; Wang, S. Electroreduction of Carbon Dioxide Driven by the Intrinsic Defects in the Carbon Plane of a Single Fe–N₄ Site. *Adv. Mater.* **2021**, *33*, 2003238.
- (11) Ju, W.; Bagger, A.; Hao, G.-P.; Varela, A. S.; Sinev, I.; Bon, V.; Roldan Cuenya, B.; Kaskel, S.; Rossmeisl, J.; Strasser, P. Understanding Activity and Selectivity of Metal-Nitrogen-Doped Carbon Catalysts for Electrochemical Reduction of CO₂. *Nat. Commun.* **2017**, *8*, 944.
- (12) Rong, X.; Wang, H.-J.; Lu, X.-L.; Si, R.; Lu, T.-B. Controlled Synthesis of a Vacancy-Defect Single-Atom Catalyst for Boosting CO₂ Electroreduction. *Angew. Chem., Int. Ed.* **2020**, *59*, 1961-1965.
- (13) Pan, Y.; Lin, R.; Chen, Y.; Liu, S.; Zhu, W.; Cao, X.; Chen, W.; Wu, K.; Cheong, W.-C.; Wang, Y.; Zheng, L.; Luo, J.; Lin, Y.; Liu, Y.; Liu, C.; Li, J.; Lu, Q.; Chen, X.; Wang, D.; Peng, Q.; Chen, C.; Li, Y. Design of Single-Atom Co–N₅ Catalytic Site: A Robust Electrocatalyst for CO₂ Reduction with Nearly 100% CO Selectivity and Remarkable Stability. *J. Am. Chem. Soc.* **2018**, *140*, 4218-4221.
- (14) Yang, H. B.; Hung, S.-F.; Liu, S.; Yuan, K.; Miao, S.; Zhang, L.; Huang, X.; Wang, H.-Y.; Cai, W.; Chen, R.; Gao, J.; Yang, X.; Chen, W.; Huang, Y.; Chen, H. M.; Li, C. M.; Zhang, T.; Liu, B. Atomically Dispersed Ni(I) as the Active Site for Electrochemical CO₂ Reduction. *Nat. Energy* **2018**, *3*, 140-147.
- (15) Zhang, Z.; Zhao, X.; Xi, S.; Zhang, L.; Chen, Z.; Zeng, Z.; Huang, M.; Yang, H.; Liu, B.; Pennycook, S. J.; Chen, P. Atomically Dispersed Cobalt Trifunctional Electrocatalysts with

Tailored Coordination Environment for Flexible Rechargeable Zn–Air Battery and Self-Driven Water Splitting. *Adv. Energy Mater.* **2020**, *10*, 2002896.

(16) Zeng, Z.; Gan, L. Y.; Bin Yang, H.; Su, X.; Gao, J.; Liu, W.; Matsumoto, H.; Gong, J.; Zhang, J.; Cai, W.; Zhang, Z.; Yan, Y.; Liu, B.; Chen, P. Orbital Coupling of Hetero-Diatomic Nickel-Iron Site for Bifunctional Electrocatalysis of CO₂ Reduction and Oxygen Evolution. *Nat. Commun.* **2021**, *12*, 4088.

(17) Liu, C.; Li, H.; Liu, F.; Chen, J.; Yu, Z.; Yuan, Z.; Wang, C.; Zheng, H.; Henkelman, G.; Wei, L.; Chen, Y. Intrinsic Activity of Metal Centers in Metal–Nitrogen–Carbon Single-Atom Catalysts for Hydrogen Peroxide Synthesis. *J. Am. Chem. Soc.* **2020**, *142*, 21861–21871.

(18) Li, F.; Han, G.-F.; Bu, Y.; Noh, H.-J.; Jeon, J.-P.; Shin, T. J.; Kim, S.-J.; Wu, Y.; Jeong, H. Y.; Fu, Z.; Lu, Y.; Baek, J.-B. Revealing Isolated M–N₃C₁ Active Sites for Efficient Collaborative Oxygen Reduction Catalysis. *Angew. Chem., Int. Ed.* **2020**, *59*, 23678–23683.

(19) Du, Z.; Chen, X.; Hu, W.; Chuang, C.; Xie, S.; Hu, A.; Yan, W.; Kong, X.; Wu, X.; Ji, H.; Wan, L.-J. Cobalt in Nitrogen-Doped Graphene as Single-Atom Catalyst for High-Sulfur Content Lithium–Sulfur Batteries. *J. Am. Chem. Soc.* **2019**, *141*, 3977–3985.

(20) Ren, W.; Tan, X.; Yang, W.; Jia, C.; Xu, S.; Wang, K.; Smith, S. C.; Zhao, C. Isolated Diatomic Ni-Fe Metal–Nitrogen Sites for Synergistic Electroreduction of CO₂. *Angew. Chem., Int. Ed.* **2019**, *58*, 6972–6976.

(21) Chen, K.; Liu, K.; An, P.; Li, H.; Lin, Y.; Hu, J.; Jia, C.; Fu, J.; Li, H.; Liu, H.; Lin, Z.; Li, W.; Li, J.; Lu, Y.-R.; Chan, T.-S.; Zhang, N.; Liu, M. Iron Phthalocyanine with Coordination Induced Electronic Localization to Boost Oxygen Reduction Reaction. *Nat. Commun.* **2020**, *11*, 4173.

(22) Wang, Y.; Tang, Y.-J.; Zhou, K. Self-Adjusting Activity Induced by Intrinsic Reaction Intermediate in Fe–N–C Single-Atom Catalysts. *J. Am. Chem. Soc.* **2019**, *141*, 14115–14119.

(23) Yang, Y.; Mao, K.; Gao, S.; Huang, H.; Xia, G.; Lin, Z.; Jiang, P.; Wang, C.; Wang, H.; Chen, Q. O-, N-Atoms-Coordinated Mn Cofactors within a Graphene Framework as Bioinspired Oxygen Reduction Reaction Electrocatalysts. *Adv. Mater.* **2018**, *30*, 1801732.

(24) Wang, X.; Wang, Y.; Sang, X.; Zheng, W.; Zhang, S.; Shuai, L.; Yang, B.; Li, Z.; Chen, J.; Lei, L.; Adli, N. M.; Leung, M. K. H.; Qiu, M.; Wu, G.; Hou, Y. Dynamic Activation of Adsorbed Intermediates via Axial Traction for the Promoted Electrochemical CO₂ Reduction. *Angew. Chem., Int. Ed.* **2021**, *60*, 4192–4198.

(25) Chen, Z.; Huang, A.; Yu, K.; Cui, T.; Zhuang, Z.; Liu, S.; Li, J.; Tu, R.; Sun, K.; Tan, X.; Zhang, J.; Liu, D.; Zhang, Y.; Jiang, P.; Pan, Y.; Chen, C.; Peng, Q.; Li, Y. Fe₁N₄–O₁ Site with Axial Fe–O Coordination for Highly Selective CO₂ Reduction over a Wide Potential Range. *Energy Environ. Sci.* **2021**, *14*, 3430–3437.

(26) Wang, Y.; Shi, R.; Shang, L.; Waterhouse, G. I. N.; Zhao, J.; Zhang, Q.; Gu, L.; Zhang, T. High-Efficiency Oxygen Reduction to Hydrogen Peroxide Catalyzed by Nickel Single-Atom Catalysts with Tetradentate N₂O₂ Coordination in a Three-Phase Flow Cell. *Angew. Chem., Int. Ed.* **2020**, *59*, 13057–13062.

(27) Huang, M.; Wang, C.; Quan, L.; Nguyen, T. H.-Y.; Zhang, H.; Jiang, Y.; Byun, G.; Ruoff, R. S. CVD Growth of Porous Graphene Foam in Film Form. *Matter* **2020**, *3*, 487–497.

(28) Zhang, Y.; Jiao, L.; Yang, W.; Xie, C.; Jiang, H.-L. Rational Fabrication of Low-Coordinate Single-Atom Ni Electrocatalysts by MOFs for Highly Selective CO₂ Reduction. *Angew. Chem., Int. Ed.* **2021**, *60*, 7607–7611.

(29) Kabir, S.; Artyushkova, K.; Serov, A.; Kiefer, B.; Atanassov, P. Binding Energy Shifts for Nitrogen-Containing Graphene-Based Electrocatalysts – Experiments And DFT Calculations. *Surf.*

Interface Anal. **2016**, *48*, 293-300.

(30) Auchi, M.; Hayashi, S.; Toyota, K.; Ukai, S. Effect of Water Vapor on the High-Temperature Oxidation of Pure Ni. *Oxid. Met.* **2012**, *78*, 51-61.

(31) Luo, L.; Su, M.; Yan, P.; Zou, L.; Schreiber, D. K.; Baer, D. R.; Zhu, Z.; Zhou, G.; Wang, Y.; Bruemmer, S. M.; Xu, Z.; Wang, C. Atomic Origins of Water-Vapour-Promoted Alloy Oxidation. *Nat. Mater.* **2018**, *17*, 514-518.

(32) Luo, L.; Li, L.; Schreiber, D. K.; He, Y.; Baer, D. R.; Bruemmer, S. M.; Wang, C. Deciphering Atomistic Mechanisms of the Gas-Solid Interfacial Reaction During Alloy Oxidation. *Sci. Adv.* **2020**, *6*, eaay8491.

(33) Xu, S.; Walter, E. D.; Zhao, Z.; Hu, M. Y.; Han, X.; Hu, J. Z.; Bao, X. Dynamic Structural Changes of SiO₂ Supported Pt–Ni Bimetallic Catalysts over Redox Treatments Revealed by NMR and EPR. *J. Phys. Chem. C* **2015**, *119*, 21219-21226.

(34) Jiang, K.; Siahrostami, S.; Zheng, T.; Hu, Y.; Hwang, S.; Stavitski, E.; Peng, Y.; Dynes, J.; Gangisetty, M.; Su, D.; Attenkofer, K.; Wang, H. Isolated Ni Single Atoms in Graphene Nanosheets for High-Performance CO₂ Reduction. *Energy Environ. Sci.* **2018**, *11*, 893-903.

(35) Cheng, Y.; Zhao, S.; Li, H.; He, S.; Veder, J.-P.; Johannessen, B.; Xiao, J.; Lu, S.; Pan, J.; Chisholm, M. F.; Yang, S.-Z.; Liu, C.; Chen, J. G.; Jiang, S. P. Unsaturated Edge-Anchored Ni Single Atoms on Porous Microwave Exfoliated Graphene Oxide for Electrochemical CO₂. *Appl. Catal. B* **2019**, *243*, 294-303.

(36) Liu, S.; Yang, H. B.; Hung, S.-F.; Ding, J.; Cai, W.; Liu, L.; Gao, J.; Li, X.; Ren, X.; Kuang, Z.; Huang, Y.; Zhang, T.; Liu, B. Elucidating the Electrocatalytic CO₂ Reduction Reaction over a Model Single-Atom Nickel Catalyst. *Angew. Chem., Int. Ed.* **2020**, *59*, 798-803.

(37) Zheng, W.; Yang, J.; Chen, H.; Hou, Y.; Wang, Q.; Gu, M.; He, F.; Xia, Y.; Xia, Z.; Li, Z.; Yang, B.; Lei, L.; Yuan, C.; He, Q.; Qiu, M.; Feng, X. Atomically Defined Undercoordinated Active Sites for Highly Efficient CO₂ Electroreduction. *Adv. Funct. Mater.* **2020**, *30*, 1907658.

(38) Li, Z.; He, D.; Yan, X.; Dai, S.; Younan, S.; Ke, Z.; Pan, X.; Xiao, X.; Wu, H.; Gu, J. Size-Dependent Nickel-Based Electrocatalysts for Selective CO₂ Reduction. *Angew. Chem., Int. Ed.* **2020**, *59*, 18572-18577.

(39) Yang, J.; Qiu, Z.; Zhao, C.; Wei, W.; Chen, W.; Li, Z.; Qu, Y.; Dong, J.; Luo, J.; Li, Z.; Wu, Y. In Situ Thermal Atomization to Convert Supported Nickel Nanoparticles into Surface-Bound Nickel Single-Atom Catalysts. *Angew. Chem., Int. Ed.* **2018**, *57*, 14095-14100.

(40) Hu, X.-M.; Hval, H. H.; Bjerglund, E. T.; Dalgaard, K. J.; Madsen, M. R.; Pohl, M.-M.; Welter, E.; Lamagni, P.; Buhl, K. B.; Bremholm, M.; Beller, M.; Pedersen, S. U.; Skrydstrup, T.; Daasbjerg, K. Selective CO₂ Reduction to CO in Water using Earth-Abundant Metal and Nitrogen-Doped Carbon Electrocatalysts. *ACS Catal.* **2018**, *8*, 6255-6264.

(41) Wang, X.; Chen, Z.; Zhao, X.; Yao, T.; Chen, W.; You, R.; Zhao, C.; Wu, G.; Wang, J.; Huang, W.; Yang, J.; Hong, X.; Wei, S.; Wu, Y.; Li, Y. Regulation of Coordination Number over Single Co Sites: Triggering the Efficient Electroreduction of CO₂. *Angew. Chem., Int. Ed.* **2018**, *57*, 1944-1948.

(42) Li, X.; Bi, W.; Chen, M.; Sun, Y.; Ju, H.; Yan, W.; Zhu, J.; Wu, X.; Chu, W.; Wu, C.; Xie, Y. Exclusive Ni–N₄ Sites Realize Near-Unity CO Selectivity for Electrochemical CO₂ Reduction. *J. Am. Chem. Soc.* **2017**, *139*, 14889-14892.

(43) Yang, X.; Cheng, J.; Fang, B.; Xuan, X.; Liu, N.; Yang, X.; Zhou, J. Single Ni Atoms with Higher Positive Charges Induced by Hydroxyls for Electrocatalytic CO₂ Reduction. *Nanoscale* **2020**, *12*, 18437-18445.

(44) Kim, C.; Jeon, H. S.; Eom, T.; Jee, M. S.; Kim, H.; Friend, C. M.; Min, B. K.; Hwang, Y.

- J. Achieving Selective and Efficient Electrocatalytic Activity for CO₂ Reduction Using Immobilized Silver Nanoparticles. *J. Am. Chem. Soc.* **2015**, *137*, 13844-13850.
- (45) Gattrell, M.; Gupta, N.; Co, A. A Review of the Aqueous Electrochemical Reduction of CO₂ to Hydrocarbons at Copper. *J. Electroanal. Chem.* **2006**, *594*, 1-19.
- (46) Li, Z.; Zeng, Q.; Ye, Z.; Zheng, W.; Sang, X.; Dong, C.-L.; Yang, B.; Pardiwala, S.; Lu, J.; Lei, L.; Wu, G.; Hou, Y. An Integrated Bioelectrochemical System Coupled CO₂ Electroreduction Device Based on Atomically Dispersed Iron Electrocatalysts. *Nano Energy* **2021**, *87*, 106187.
- (47) Zhao, C.; Dai, X.; Yao, T.; Chen, W.; Wang, X.; Wang, J.; Yang, J.; Wei, S.; Wu, Y.; Li, Y. Ionic Exchange of Metal–Organic Frameworks to Access Single Nickel Sites for Efficient Electroreduction of CO₂. *J. Am. Chem. Soc.* **2017**, *139*, 8078-8081.
- (48) Su, P.; Iwase, K.; Nakanishi, S.; Hashimoto, K.; Kamiya, K. Nickel-Nitrogen-Modified Graphene: An Efficient Electrocatalyst for the Reduction of Carbon Dioxide to Carbon Monoxide. *Small* **2016**, *12*, 6083-6089.
- (49) Jiang, K.; Siahrostami, S.; Akey, A. J.; Li, Y.; Lu, Z.; Lattimer, J.; Hu, Y.; Stokes, C.; Gangishetty, M.; Chen, G.; Zhou, Y.; Hill, W.; Cai, W.-B.; Bell, D.; Chan, K.; Nørskov, J. K.; Cui, Y.; Wang, H. Transition-Metal Single Atoms in a Graphene Shell as Active Centers for Highly Efficient Artificial Photosynthesis. *Chem* **2017**, *3*, 950-960.
- (50) Liu, C.; Wu, Y.; Sun, K.; Fang, J.; Huang, A.; Pan, Y.; Cheong, W.-C.; Zhuang, Z.; Zhuang, Z.; Yuan, Q.; Xin, H. L.; Zhang, C.; Zhang, J.; Xiao, H.; Chen, C.; Li, Y. Constructing FeN₄/Graphitic Nitrogen Atomic Interface for High-Efficiency Electrochemical CO₂ Reduction over a Broad Potential Window. *Chem* **2021**, *7*, 1297-1307.
- (51) Goodpaster, J. D.; Bell, A. T.; Head-Gordon, M. Identification of Possible Pathways for C–C Bond Formation during Electrochemical Reduction of CO₂: New Theoretical Insights from an Improved Electrochemical Model. *J. Phys. Chem. Lett.* **2016**, *7*, 1471-1477.
- (52) Sundararaman, R.; Goddard III, W. A.; Arias, T. A. Grand Canonical Electronic Density-Functional Theory: Algorithms and Applications to Electrochemistry. *J. Chem. Phys.* **2017**, *146*, 114104.
- (53) Kim, D.; Shi, J.; Liu, Y. Substantial Impact of Charge on Electrochemical Reactions of Two-Dimensional Materials. *J. Am. Chem. Soc.* **2018**, *140*, 9127-9131.
- (54) Nørskov, J. K.; Rossmeisl, J.; Logadottir, A.; Lindqvist, L.; Kitchin, J. R.; Bligaard, T.; Jónsson, H. Origin of the Overpotential for Oxygen Reduction at a Fuel-Cell Cathode. *J. Phys. Chem. B* **2004**, *108*, 17886-17892.
- (55) Calle-Vallejo, F.; Koper, M. T. M. Theoretical Considerations on the Electroreduction of CO to C₂ Species on Cu(100) Electrodes. *Angew. Chem., Int. Ed.* **2013**, *52*, 7282-7285.
- (56) Peterson, A. A.; Abild-Pedersen, F.; Studt, F.; Rossmeisl, J.; Nørskov, J. K. How Copper Catalyzes the Electroreduction of Carbon Dioxide into Hydrocarbon Fuels. *Energy Environ. Sci.* **2010**, *3*, 1311-1315.
- (57) Wang, X.; Niu, H.; Liu, Y.; Shao, C.; Robertson, J.; Zhang, Z.; Guo, Y. Theoretical Investigation on Graphene-Supported Single-Atom Catalysts for Electrochemical CO₂ Reduction. *Catal. Sci. Technol.* **2020**, *10*, 8465-8472.

For Table of Contents Only

